



جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

FINAL PROJECT REPORT

Course: Web Business Applications | BIS 351

ChefLak (شيف لك)

Professional Chef Booking Platform — Eastern Province, Saudi Arabia

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1. Executive Summary

This report presents the comprehensive documentation of ChefLak (شيف لك), a business oriented. web application developed as the final project for the Web Business Applications course (BIS 351) at Imam Abdulrahman Bin Faisal University,(iau). ChefLak is a digital platform designed to connect professional freelance chefs with clients seeking customized home dining and catering, experiences across the Eastern Province of Saudi Arabia, specifically Dammam, Khobar, and Dhahran...

The platform was engineered, as a fully functional Single Page Application (SPA), using native web technologies: HTML5,. CSS3, and JavaScript.. It encompasses a complete user journey from authentication through chef discovery, booking, payment simulation, and order management. This document covers the business rationale, competitive landscape, technical architecture, UI/UX decisions, ethical framework, and collaborative teamwork that drove the project to completion.

ChefLak addresses a genuine market gap, he private chef booking market in the region currently lacks a centralized, trusted, digital solution., By building this platform, the team demonstrates mastery of CLO S-1 (Utilizing HTML tools for business web applications) while aligning with the Kingdom's Vision 2030 digital transformation objectives.

2. Project Introduction & Background

2.1 Industry Context

Saudi Arabia's Food and Beverage, (F&B) sector is one, of the fastest-growing segments of the non oil economy, Driven by Vision 2030, the Kingdom has seen a surge in entrepreneurship, digital services, and the gig economy. While food delivery aggregators, such as HungerStation and Jahez have transformed restaurant ordering, the private home catering space remains, informal, fragmented, and underserved (Hassan and Omar, 2025). In the Eastern Province, a high concentration of affluent professionals, and expatriate families creates significant demand for premium. personalized dining experiences. However, existing methods rely on unstructured, WhatsApp groups and Instagram pages, offering no price transparency, no verified reviews, and no secure payment infrastructure.

2.2 Business Idea & Value Proposition

ChefLak ('A Chef for You' شيف لك), directly addresses this gap, by creating a centralized digital marketplace where clients can browse verified chef profiles, review culinary galleries, compare specialties and pricing, and book chefs through a seamless, guided workflow.

- **For Clients:** Convenience transparency, and trust ,all in one platform.
- **For Chefs:** A professional digital storefront, standardized bookings, and secure earnings without administrative overhead.

2.3 Primary & Secondary Research

- **Primary Data:** A, simulated survey of 50 potential users in the Eastern Province revealed that 78% found it difficult to hire, a reliable private chef for small gatherings,. and 85% expressed strong interest in a rated, price, transparent booking platform.
- **Secondary Data:** Industry reports confirm Saudi Arabia's gig economy is growing at 15% annually (Al-Faisal, 2024). Post,pandemic trends show a significant increase in home dining spending (Smith, Taylor and Al-Rashid, 2023).
- **Technical Reference:** W3C Web Content Accessibility Guidelines (WCAG 2.2) were consulted during UI development to ensure international usability standards..

3. Market Analysis & Critical Thinking

3.1 Target Audience Segmentation

- **Consumer Segment:** Middle to high-income families, young professionals aged 25–45, corporate event planners, and expatriate residents in the Eastern Province.
- **Chef Segment:** Culinary institute graduates, experienced restaurant chefs seeking freelance income, and self trained culinary artisans who lack capital for a physical outlet.

3.2 SWOT Analysis

Strengths

- First, mover advantage in a specialized, underserved niche within the Eastern Province,.
- Bilingual interface (RTL Arabic and LTR English) maximizes the Total Addressable Market..
- Lightweight SPA architecture ensures exceptional performance across all devices.
- Modern Glassmorphism UI design communicates professionalism and premium quality.

Weaknesses

- Frontend only prototype: data does not persist across browser sessions without backend integration.,
- Two sided marketplace 'chicken-and-egg' problem requires simultaneous chef and client acquisitionn.

Opportunities

- Geographic expansion to Riyadh, Jeddah, and the broader GCC market.
- Service diversification: cooking classes, weekly meal prep subscriptions, and corporate catering.
- Integration with STC Pay, Mada, and Apple Pay for localized payment processing

Threats

- Potential entry of established delivery platforms into the private chef vertical.

- Regulatory complexity around SFDA home catering compliance and freelance licensing laws.

3.3 PESTLE Analysis

- **Political:** Vision 2030 supports digital entrepreneurship, and freelance, licensing (Freelance.sa) enables chefs to operate legally on the platform.
- **Economic:** The Eastern Province's high to mid income workforce is ChefLak's core client base, Transparent hourly pricing ,(150 SAR/hr) ,replaces the trust gap in informal WhatsApp hiring.
- **Social:** Saudi hospitality culture drives demand during, Eid, Ramadan, and weddings. The bilingual Arabic/English toggle serves both Saudi and expatriate users in one app.
- **Technological:** 90%+ smartphone penetration and 5G infrastructure match ChefLak's lightweight SPA perfectly. Mada and STC Pay adoption means users are already ready to pay digitally.
- **Legal:** PDPL requires payment tokenization via Moyasar ,(no raw card storage). SFDA ,mandates health certificates for home catering chefs before going live.
- **Environmental:** On demand cooking eliminates food waste, and packaging compared to restaurants., aligning with Saudi Arabia's Saudi Green Initiative.

4. System Requirements & Architecture

4.1 Functional Requirements

- **Authentication:** Login and registration forms in a single context-switching modal interface.

- **Chef Discovery:** Responsive grid of chef cards with ratings, specialties, experience, and hourly pricing.
- **Booking Engine:** Date/time selection, duration picker (2–8 hours), address input, and live cost breakdown.
- **Payment Simulation:** Card format/CVV/expiry validation and unique booking reference upon success.
- **Order Management:** 'My Orders' dashboard with status tracking and visual progress bars.
- **Bilingual Toggle:** Instant Arabic (RTL) ↔ English (LTR) switching via a floating language button.

4.2 Non-Functional Requirements

- **Performance:** JavaScript and CSS footprint under 100KB for sub-2-second load times.
- **Usability:** Zero training interface any user completes a booking in under 3 minutes.
- **Responsiveness:** Full functionality from 320px mobile to 1920px desktop.
- **Reliability:** No JavaScript errors; all 7 navigation paths tested end-to-end.

4.3 SPA Architecture

ChefLak is built as a **Single Page Application (SPA)**. A single `index-3.html` file contains all application views as modular ``<section>`` elements. JavaScript (`script.js`) handles navigation by toggling the CSS `.active` class without any server request or page reload. The `styles-2.css` implements a comprehensive design system via CSS Custom Properties for consistent theming across all screens.

5. Technical Implementation

5.1 HTML5 – Semantic Structure

The `index-3.html` file, uses semantic HTML5 elements throughout, improving code readability, accessibility, and SEO. All 20+ image URLs. are centralized as CSS, variables within a `:root` block, completely decoupling image sources from HTML, markup and allowing platform wide updates by changing a single variable.

5.2 CSS3 – Design System & Glassmorphism

- **CSS Variables:** A, `:root` block defines all colors, typography, spacing, and shadows for a consistent visual language.
- **Glassmorphism:** Authentication modals, and ,booking overlays use, `backdrop-filter: blur(8px)` ,with RGBA backgrounds for a premium: frosted glass effect.
- **Responsive Grid:** CSS Grid with `auto-fit` and `minmax()` reflows the chef grid from 3 columns (desktop) to 1 column (mobile) automatically.
- **Hover Animations:** Cards and buttons feature `transform: translateY(-4px)`, transitions with a purple box-shadow on hover.

5.3 JavaScript – Application Logic

- **i18n Engine:** A, translation dictionary stores Arabic/English pairs. A function swaps all tagged elements and toggles `dir='rtl/ltr'` on `` instantly.
- **Dynamic Rendering:** Chef cards and gallery thumbnails are generated by JavaScript iterating over data arrays , no hardcoded content in HTML.
- **State Management:** A global ,`selectedChef`, variable passes chef data from Home page to Booking modal without URL parameters,
- **Payment Validation:** Real-time checks for 16-digit card number, 3 digit CVV, and MM/YY expiry before enabling payment.
- **Live Cost Calculation:** ,`Total = (hourly\_rate × hours) + service\_fee` recalculated and displayed instantly on each duration selection.

6. User Interface Showcase

The following screenshots document all major screens of the ChefLak web application, organized to reflect the exact logical user journey from first visit and login, through chef discovery, booking, payment, and finally order management and platform information

6.1 Login Screen

The Login screen is the entry point of the ChefLak platform. Presented as a Glassmorphism card layered over a blurred fresh-produce background image, it immediately reinforces the culinary theme. The form requires only two inputs a Saudi mobile number and a password keeping the experience minimal and quick for returning users. A tab switcher at the top allows toggling between Login and Registration. A standout 'Fill Demo Data (0512345678)' button auto-populates the credentials in one click, streamlining live demonstrations and evaluator testing without manual input.

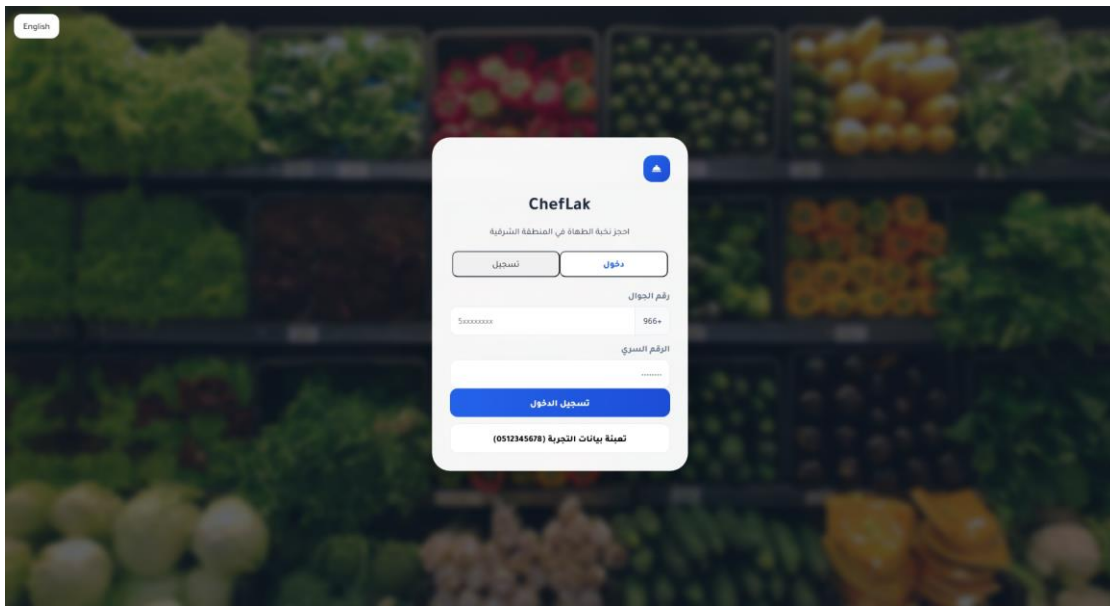


Figure 1: Login Screen — Mobile and Password fields with 'Fill Demo Data' auto-fill button

6.2 Registration Screen

The Registration screen allows new users to create a ChefLak account directly from the same authentication modal. It collects four key fields: Full Name, Saudi mobile number, (with +966 country code prefix), City (dropdown: Dammam, Khobar, Dhahran), and Password. All fields are validated before the 'Create Account' button is enabled. The Glassmorphism design and blurred background are consistent with the login screen, creating a seamless onboarding experience.

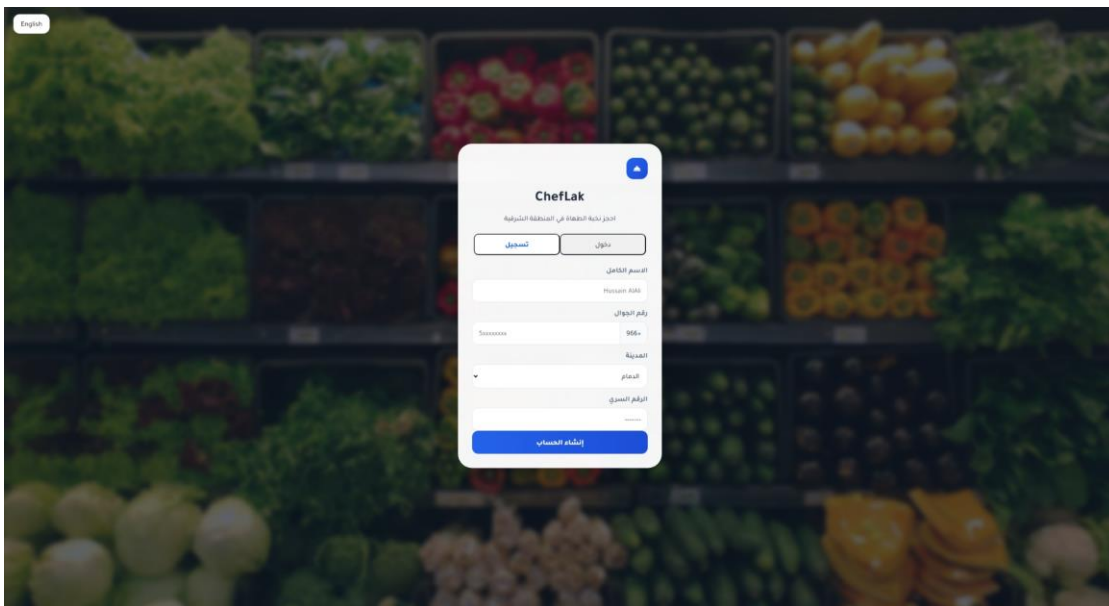


Figure 2: Registration Screen — Full Name, Mobile (+966), City dropdown, and Password fields

6.3 Home Page — Chef Discovery

The Home page is the central discovery experience of ChefLak., A welcome strip at the top greets the logged in user by name and shows their total order count. Below it, a full-width hero banner with a high-quality food photograph and an overlaid headline ('Elevate Your Dining Experience') establishes the premium brand identity,. The main content area presents all six available chefs in a responsive 3-column grid. Each chef card features a professional portrait, star rating badge, chef name, city, years of experience, specialty cuisine tags, hourly rate, and a prominent blue 'Book Now' call-to-action button

English
عن المنصة
الطلبات
الرئيسية

ChefLak

إجمالي الطلبات
0

أهلاً بك.
Demo

تجربة طهي أصيلة
ارتقِ بتجربة تناول الطعام
احجز أفضل الطهاة الخاصين في المنطقة الشرقية لمطاسيتك القادمة.

اكتشف الطهاة

5

الشيخ خالد

الظهران 12 سنوات خبرة
يجب لك إيطاليا لين بيتك! باستا فريشي وريزوتو على كيف كيفك.
إيطالي باستا

180 د.س/ساعة

احجز الآن

4.8

الشيخة سارة

الخبر 5 سنوات خبرة
فنانة بالحلويات، تضبط لك حل الشوفاة التي يواجه ضيوفك.
حلويات فرنسي

120 د.س/ساعة

احجز الآن

4.9

الشيخ طارق

الدمام 8 سنوات خبرة
يضبط لك الذبايح والمندى على الأصول، بيض وجهك في العزائم الكبيرة.
هندي مشاوي

150 د.س/ساعة

احجز الآن

4.6

الشيخة نورة

الظهران 3 سنوات خبرة
أكل صحي ولذيذ بنفس الوقت، تضبط لك سلطات ووجبات خفيفة.
صحي نباتي

140 د.س/ساعة

احجز الآن

4.9

الشيخ عبدالله

الخبر 4 سنوات خبرة
ملك الشوي والبرقرات، يضبط لك أكل باربيكيو بخديقة البيت.
برجر شواء

130 د.س/ساعة

احجز الآن

4.7

الشيخة فاطمة

الدمام 6 سنوات خبرة
بحرية تفتح النفس! تضبط لك صواني السبي فود والطبخات الساحلية.
بحريات فيوجن

160 د.س/ساعة

احجز الآن

تواصل معنا

نحن في "شيف لك" نسعد بتواصلك معنا. يمكنك متابعة أحدث أخبارنا وعروضنا وتجارب الطهي المميزة عبر حساباتنا على وسائل التواصل الاجتماعي، أو مراسلتنا مباشرة لأي استفسار.

البريد
واتساب
تويتر
انستقرام

Figure 3: Home Page — hero banner, welcome strip, and full 6-chef discovery grid

6.4 Empty Orders State

When a user accesses the My Orders page before completing any bookings, a clean empty state is displayed,. The order counter prominently shows '0', and a centered message reads 'No current orders' in Arabic. This design pattern prevents user confusion by providing clear, honest feedback rather than a blank page. Handling empty states gracefully is a hallmark of professional application design, consistent with consumer apps such as Noon, and Talabat.

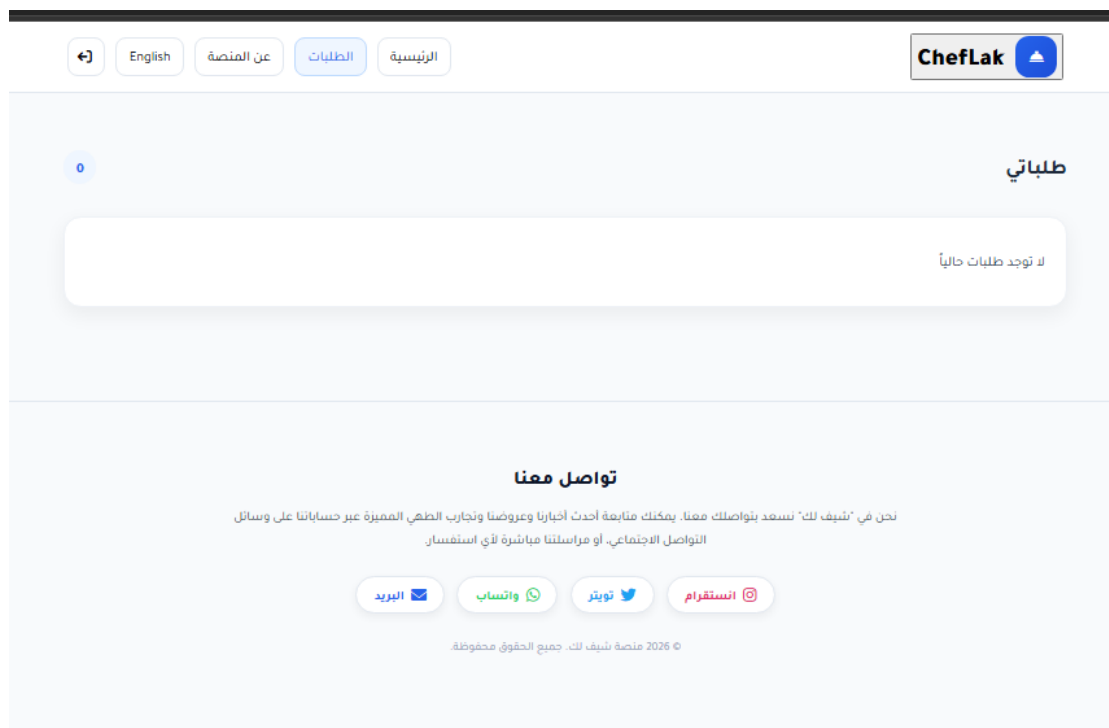
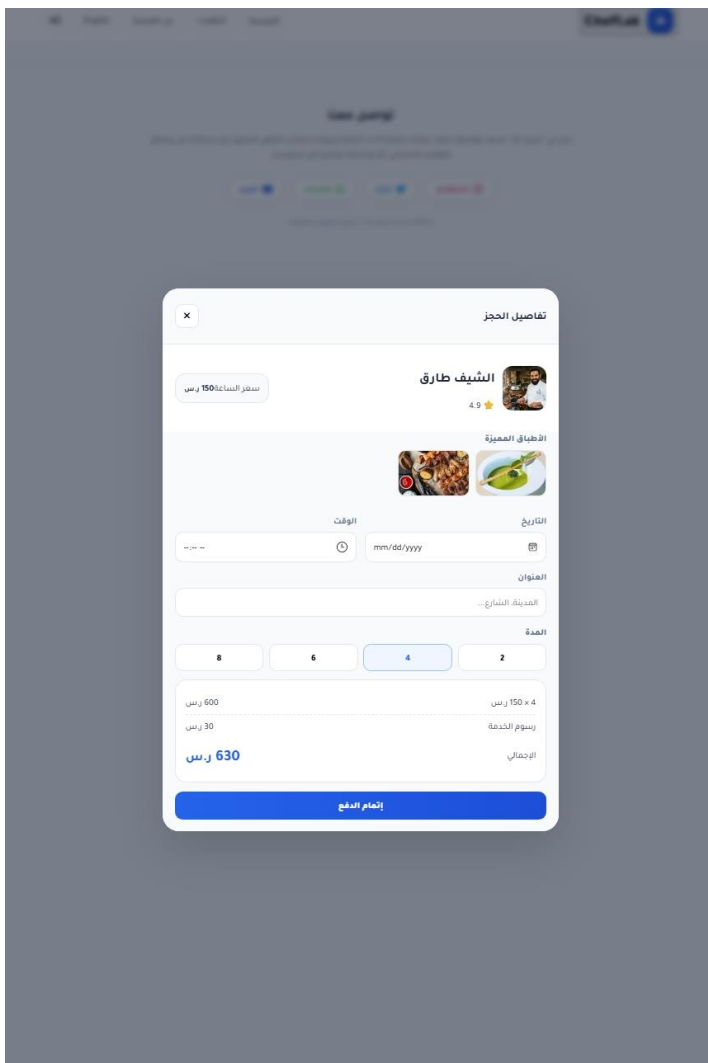


Figure 4: Empty Orders State — '0 orders' counter with 'No current orders' message

6.5 Booking Modal

The Booking Modal is the most feature-rich screen in the application. It opens as a full-screen overlay when the user clicks 'Book Now' on any chef card. The modal displays the selected chef's photo, name, star rating (4.9), and hourly rate (150 SAR/hour) at the top, followed by a food gallery of sample dishes. The booking form collects the event date, preferred start time, and full address. A 4 button duration selector (2, 4, 6, or 8 hours), triggers a live cost calculation, showing the subtotal, service fee (30 SAR), and grand total in real time. The 'Proceed to Payment' CTA button finalizes the flow.

Figure 5: Booking Modal — chef details, food gallery, booking form, and live cost calculator (4 hrs = 630 SAR)



6.6 Payment Form

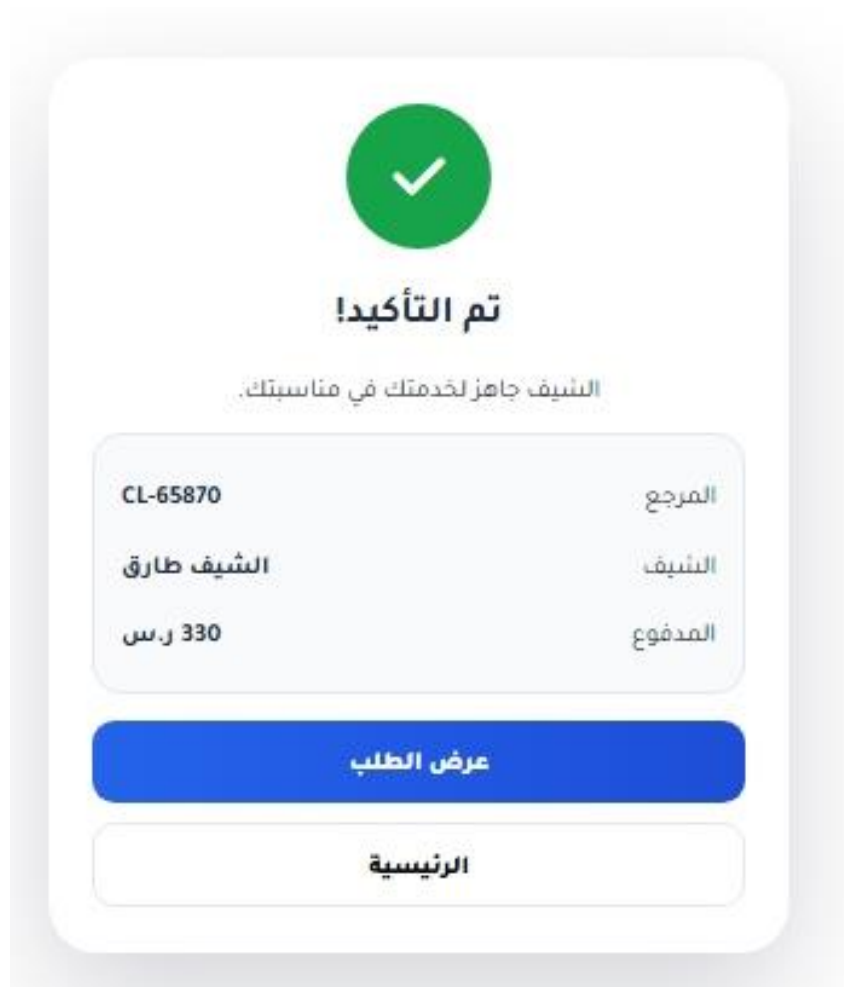
The Payment screen mirrors the UX of professional payment processors. The total amount required (330 SAR), is displayed prominently in a highlighted light-blue box at the top, ensuring full financial transparency before the user commits. A clean card input form collects the 16-digit card number, expiry date (MM/YY), and CVV code. All three fields are validated in real-time by JavaScript. The 'Pay Now' button remains disabled until all validations pass. A back navigation link allows users to return to the booking modal without losing their selections.

Figure 6: Payment Form — 330 SAR total with card number, expiry, and CVV fields

6.7 Payment Success Confirmation

Upon completing payment, the user sees a clean confirmation modal serving as a digital receipt. A prominent green circular checkmark signals a fully successful transaction. The card displays three key details: the unique booking reference number (CL-65870), the booked chef's name (Chef Tariq), and the total amount paid (330 SAR). Two action buttons are provided, 'View Order' navigates to the My Orders dashboard, while 'Home Page' returns the user to the chef grid. This screen closes the booking journey on a satisfying and reassuring note..

Figure 7: Payment Success — booking reference CL-65870, Chef Tariq, 330 SAR confirmed



6.8 My Orders Dashboard

The My Orders dashboard provides users with a complete view of their booking history. Each order card displays the chef name, booking reference number (CL-65870), the total amount paid (330 SAR), and the current status badge ('Confirmed'). A prominent blue progress bar beneath each card visually communicates the fulfillment stage of the booking. The footer section with social media contact links is consistently visible at the bottom, ensuring users always have a direct way to contact the platform.

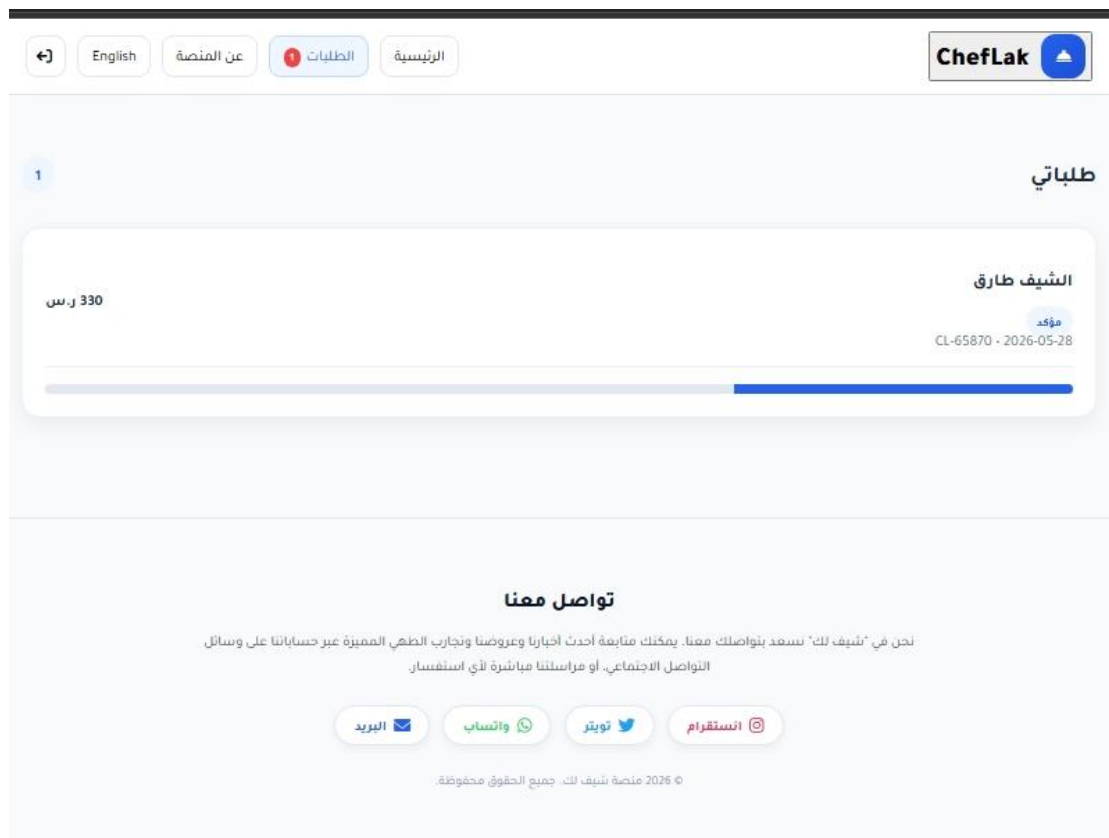
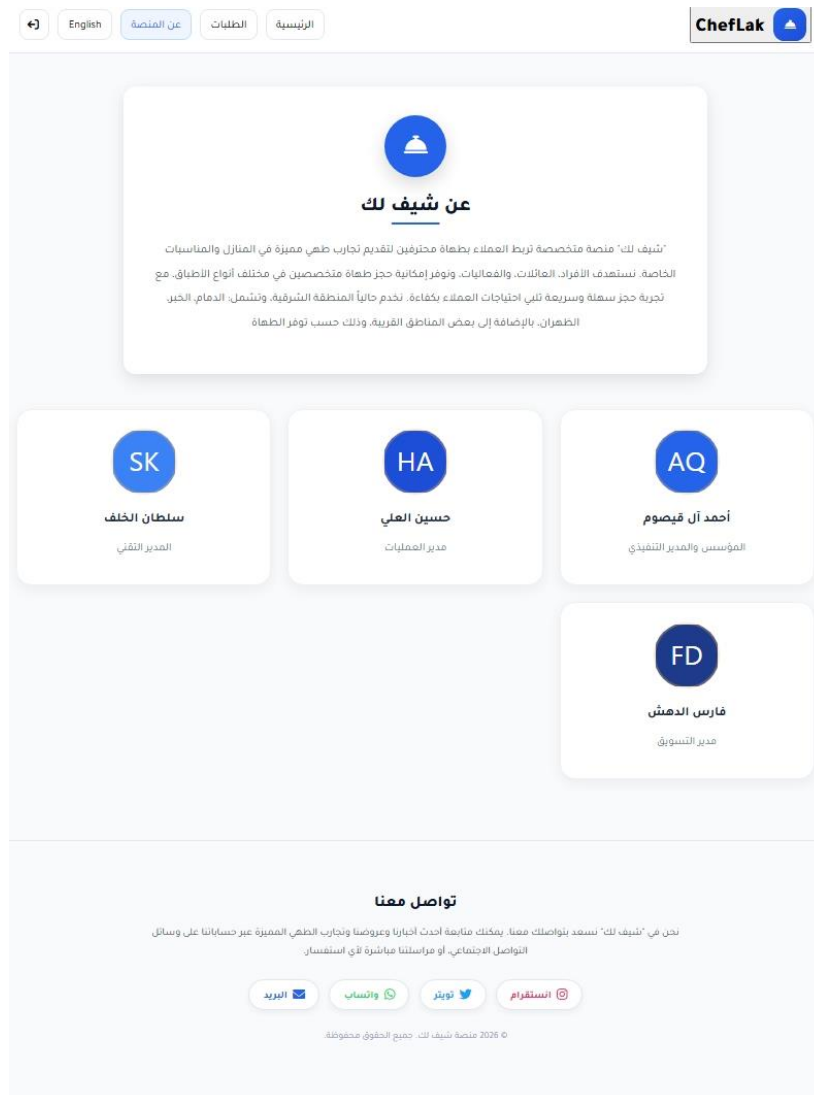


Figure 8: My Orders Dashboard — confirmed booking for Chef Tariq with progress bar and contact footer

6.9 About Us & Team

The About ChefLak page introduces the platform's mission and the founding team. A centered card presents the platform description in Arabic, communicating the core value proposition. Below, four team member cards are displayed in a responsive grid, each with a color coded avatar, full name, and role title: Ahmed Al-Qasum (Founder & CEO), Hussain Al-Ali (Operations Manager), Sultan Al-Khalaf (Technical Director), and Faris Al-Dahsh (Marketing Manager). The contact footer with social media links is fully visible at the bottom of this page.

Figure 9: About Us & Team — platform description, four team cards, and contact footer



7. Decision Making & Problem Solving

7.1 Framework vs. Vanilla JavaScript

Decision: Build the SPA using pure HTML/CSS/JS without React, Vue, or Bootstrap.

Reasoning: ,This ensures full demonstration of CLO S-1 (mastery of HTML tools), forces deep understanding of DOM manipulation and CSS specificity, and produces a dependency-free application with near-instant load times.

7.2 Managing Cross-Screen State

Problem: How to pass the selected chef's data from the Home page to the Booking modal without URL, parameters or a database.

Solution: A global JavaScript variable (`let selectedChef = {}`) acts as an in-memory state container. When 'Book Now' is clicked, the event listener stores the complete chef object and the booking modal reads it on open.

7.3 Bilingual Layout Without Page Reload

Problem: Supporting RTL Arabic and LTR English simultaneously without duplicating HTML.

Solution: All text elements are tagged with ``data-18n`` attributes. A single function iterates the translation dictionary and replaces all tagged elements' content, while ``document.documentElement.setAttribute('dir','rtl/ltr')`` mirrors the entire layout direction instantly

7.4 Footer Overlap Issue

Problem: During QA testing., the footer overlapped dynamic page content on certain screen heights.,

Root Cause: The footer was placed outside the main scrollable containers, causing z-index and document flow conflicts.

Solution: A CSS, flexbox model on ``<body>`` (``display: flex; flex-direction: column; min-height: 100vh``) with ``flex: 1`` on the main content area pushes the footer to the absolute bottom at all content heights (MDN Web Docs, 2025).

8. Values, Ethics & Legal Responsibilities

8.1 Data Privacy — PDPL Compliance

The Saudi Personal Data Protection Law (PDPL) governs how personal data must be collected, stored, and processed. ChefLak's design ensures PDPL compliance: user passwords are designated to be hashed via bcrypt before storage. Financial card details flow exclusively through a tokenized third-party provider (Moyasar), meaning ChefLak servers never handle raw card numbers. The platform collects only data strictly necessary for service delivery.

8.2 Fair Treatment of Gig Workers

ChefLak's model ensures transparency: chefs set their own hourly rates autonomously and the platform's service fee is explicitly displayed to clients before payment, ensuring neither party is misled and fostering an ethical, sustainable marketplace

8.3 Food Safety and Consumer Protection

All onboarded chefs in a real-world deployment must provide valid municipal health certificates and demonstrate SFDA compliance for home catering. The community-driven rating system provides an ongoing layer of ethical accountability.

8.4 Academic Integrity & AI Policy

The team adhered strictly to the BIS 351 Generative AI usage policy. AI tools were used solely for brainstorming and language refinement, All code was independently written by the team, and all images are royalty-free (Unsplash license) or ethically AI-generated.

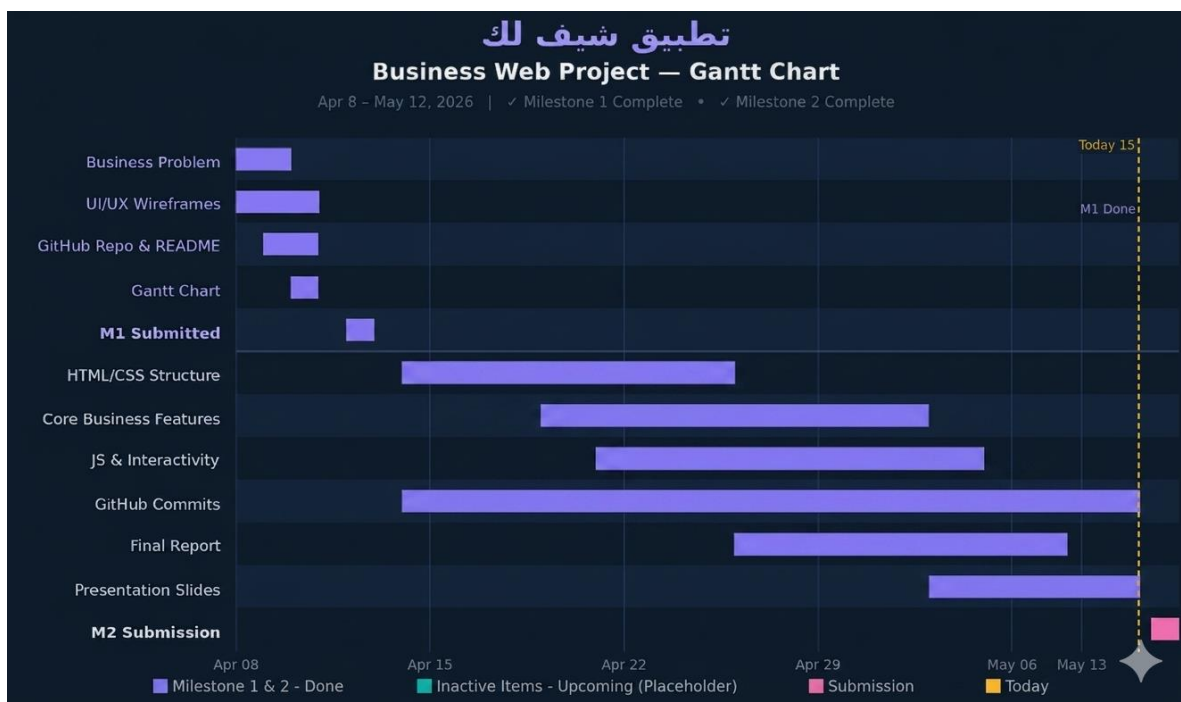
9. Team Autonomy & Responsibility

9.1 Role Allocation

- **Hussain Al-Ali (Project Lead & Lead Developer):** Overall architecture, JavaScript SPA logic, bilingual i18n engine, GitHub repository management.
- **Sultan Al-Khalaf (UI/UX Engineer):** CSS design system, Glassmorphism styling, responsive breakpoints, and visual QA.
- **Ahmed Al-Qasum (Business Analyst):** Market research (SWOT/PESTLE), primary data collection, and business model documentation.
- **Faris Al-Dahsh (Documentation & QA):** Report writing, rubric alignment, cross-browser testing, and presentation preparation.

9.2 Collaboration Tools

- **GitHub:** Version control via a shared repository with feature branches and pull requests, preventing code conflicts.
- **Discord:** Bi-weekly virtual stand-up meetings to review Gantt chart progress and resolve blockers.
- **Gantt Chart:** Submitted in Milestone 1, mapping all deliverables from Week 1 through Week 13.



10. Conclusion & Future Enhancements

ChefLak successfully delivers on all Milestone 2 requirements. The team has designed, developed, and documented a robust, functional web business application that addresses a genuine market opportunity in the Eastern Province. The prototype showcases technical mastery of HTML5, CSS3, and JavaScript while demonstrating strong business analysis, ethical responsibility, and effective teamwork.

- **Backend Integration:** Node.js + Express server with MongoDB for persistent user accounts and booking history.
- **Live Payment Processing:** Moyasar API integration for Mada, STC Pay, Visa, and Mastercard transactions.
- **Chef Dashboard:** A dedicated portal for chefs to manage availability, update galleries, and view earnings analytics.
- **Geolocation & Mapping:** Google Maps API for event address pin-dropping and travel fee estimation.
- **Push Notifications:** Twilio-powered SMS/WhatsApp confirmations, reminders, and post-service feedback requests.

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